Regression Analysis of Racial Bias in Jury Selection

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Introduction:

In 1986, the Supreme Court Case Baston v Kentucky ruled it unconstitutional for prosecutors to use race as a reason for exercising peremptory strikes against African-American potential jurors. To investigate whether racial bias persists in jury selection, APM Reports compiled data on 418 trial proceedings of Mississippi's Fifth Circuit Court District overseen by District Attorney Doug Evans or his assistants. The researchers reviewed public records and conducted a page-by-page analysis of handwritten docket books across eight courthouses in the district, covering trials from 1992 to 2017. For each trial, they created a digital record noting jurors called for duty, peremptory strikes by both prosecution and defense, juror race, the final jury list, and voir dire transcripts.

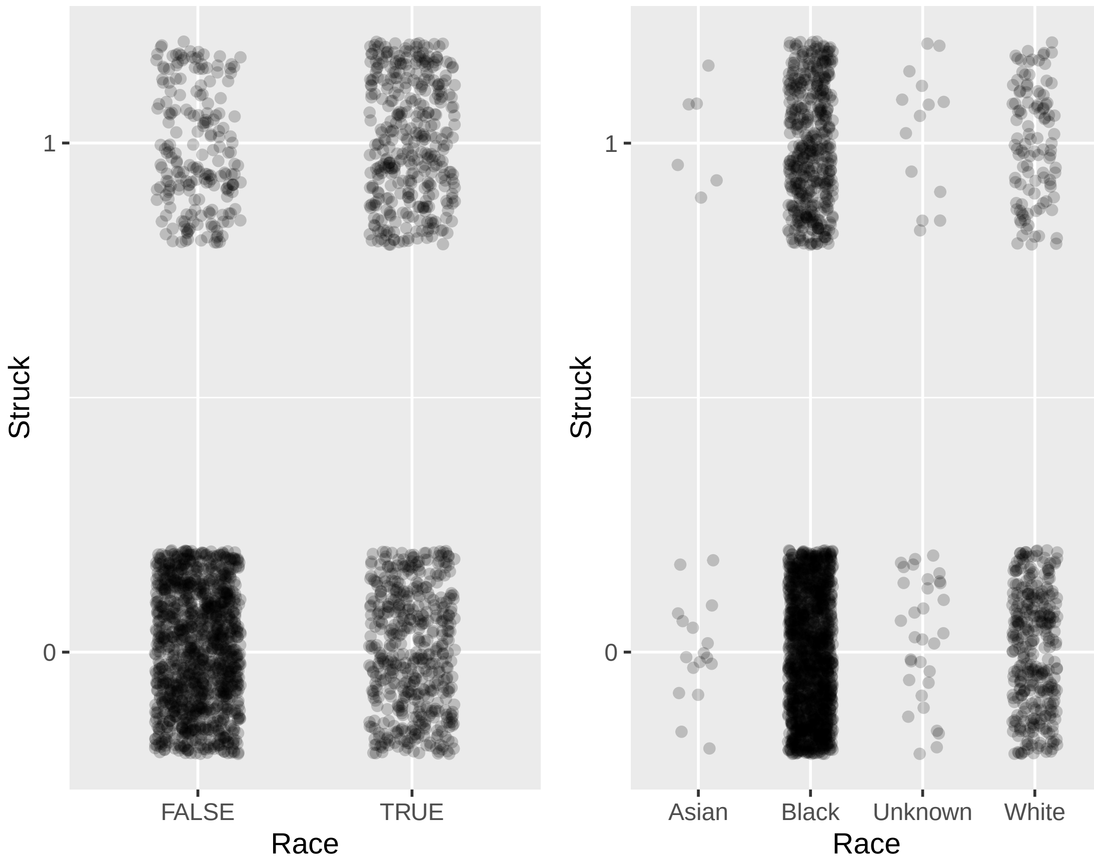
In this study, we use the APM dataset to examine whether the race of the defendant or the racial match between juror and defendant better predicts the odds of a juror being struck. We also explore whether the likelihood of Black jurors being struck varies depending on other variables collected in the dataset.

Results:

In exploring whether a juror sharing the same race as the defendant is a stronger predictor of being struck from the jury than the defendant's race alone, we created two models. One model included whether the juror and defendant were of the same race, and the other used only the race of the defendant. We then plotted each variable against the proportion of jurors struck.

Figure 1 displays juror strike outcomes based on two different racial factors. The left panel shows strike rates depending on whether the juror and the defendant are of the same race (same_race), while the right panel shows strike rates based on the defendant's race (defendant_race). In the left panel, we can clearly see that jurors who are not the same race as the defendant (same_race = FALSE) are more frequently struck than those who share the defendant's race (same_race = TRUE). In contrast, the right panel shows no strong or consistent pattern in strike rates across different racial categories of the defendant. The differences between racial groups like Black, White, Asian, or Unknown are not as visually distinct. Together, these visuals suggest that the relative racial relationship between the juror and the defendant provides a stronger explanation of strike patterns than the defendant's race alone.

Figure 1: Juror Strikes by Racial Factors

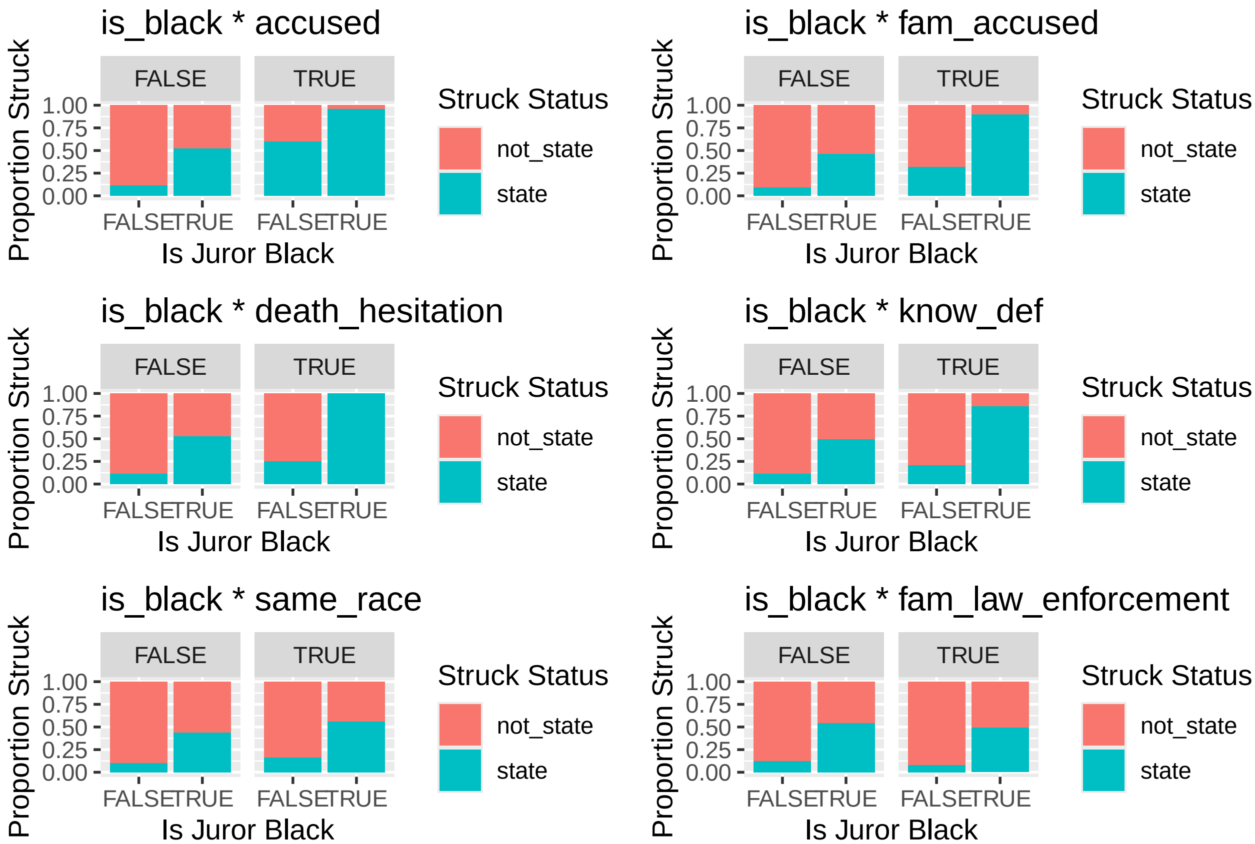


Following our visual analysis, we conducted a Likelihood Ratio Test (LRT) and compared AIC values to support our findings. The LRT for the model including same_race produced a p-value less than 0.05, indicating that it provides a significantly better fit than the model without it. In contrast, the LRT for the model including defendant_race yielded a p-value greater than 0.05, suggesting that it does not improve model fit. The AIC values support this finding, further indicating that same_race is a better predictor than defendant_race for modeling whether a juror was struck.

To examine whether there are statistically significant interactions between the variable is_black and any of the six predictors in Group A, we constructed six separate logistic regression models, each including an interaction term between is_black and one of the predictors. Before performing model comparisons, we plotted Pearson residuals versus fitted values for each model to assess model fit. None of the plots indicated violations of the linearity assumption of logistic regression. We therefore proceeded with further graphical and statistical analyses.

Figure 2 presents the fitted models and their corresponding interaction plots. The graphs show no evidence of interaction between is_black and the variables accused, same_race, and fam_law_enforcement, as the difference in the proportion of jurors struck between Black and non-Black jurors remains relatively consistent across the levels of each of these variables. In contrast, the variables fam_accused, death_hesitation, and know_def display clear signs of interaction. The difference in strike rates between Black and non-Black jurors is larger when jurors have a family connection to the accused, express hesitation about the death penalty, or personally know the defendant.

Figure 2



To support the visual analysis, we conducted two model comparison tests for each variable, a likelihood ratio test (LRT) comparing the interaction model to the corresponding non-interaction model, and an AIC comparison between the same models. The LRT results confirmed the patterns observed in the plots. Specifically, the p-values for the interaction models involving fam_accused, death_hesitation, and know_def were all below 0.05, indicating that the inderaction models provide a better fit than the non-interaction model. In contrast, the LRTs for accused, same_race, and fam_law_enforcement yielded p-values above 0.05, suggesting no significant improvement in model fit after adding an interaction. The AIC results mainly aligned with these findings, the interaction models for fam_accused, death_hesitation, and know_def had lower AIC values than the non-interaction model, indicating better model fit. The only exception was the same_race model, for which the AIC was slightly lower in the interaction model. However, since the AIC difference was minimal, this is not significant.

Discussion:

Our analysis of strikes in jury selection within the Mississippi Court District overseen by Doug Evans indicates that sharing the race of the defendant is a better predictor of the odds of being struck than the defendant's race alone.

Furthermore, our interaction models reveal that the odds of Black jurors being struck were influenced by factors such as having family ties to the accused, expressing hesitation about the death penalty, or knowing the defendant personally.

However, our analysis is limited in its scope to trials overseen by Doug Evans within one Mississippi

Court District. Therefore, these findings cannot be generalized to jury selection patterns across other parts of the United States. While we identify patterns within this dataset, we cannot conclude that the same factors apply to all trials nationwide. Despite this limitation, our study adds to a growing body of evidence suggesting that racial disparities in jury selection persist as a systemic issue. In future work, we hope to expand our analysis to other jurisdictions in the U.S. to assess whether the same interaction effects hold and racial disparity between juror and defendant is a stronger predictor of being struck than race alone.

R Appendix:

```
apm <- read.csv("http://people.carleton.edu/~apoppick/ClassData/apm.csv", header = TRUE)
```

```
large_model <- glm( struck_state_bin ~ accused + is_black + fam_accused + death_hesitation + kn
small_model <- glm(struck_state_bin ~ accused + is_black + fam_accused + death_hesitation + kno
anova(large_model, small_model, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: struck_state_bin ~ accused + is_black + fam_accused + death_hesitation +
##     know_def + same_race + fam_law_enforcement + know_vic + know_wit +
##     fam_crime_victim + crime_victim + prior_info + gender
## Model 2: struck_state_bin ~ accused + is_black + fam_accused + death_hesitation +
##     know_def + same_race + fam_law_enforcement
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      2280     1880.7
## 2      2287     1887.5 -7  -6.8792   0.4416
```

```
apm_glm <- glm(struck_state_bin ~ accused + is_black + fam_accused + death_hesitation + know_de
summary(apm_glm)
```

```
##
## Call:
## glm(formula = struck_state_bin ~ accused + is_black + fam_accused +
##     death_hesitation + know_def + same_race + fam_law_enforcement,
##     family = "binomial", data = apm)
##
## Coefficients:
##                         Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -2.4307     0.1012 -24.017  < 2e-16 ***
## accusedTRUE               2.5128     0.5455   4.606 4.10e-06 ***
## is_blackTRUE              1.8972     0.1411  13.443  < 2e-16 ***
## fam_accusedTRUE           1.8476     0.1620  11.402  < 2e-16 ***
## death_hesitationTRUE      1.8243     0.5916   3.084 0.002044 **
## know_defTRUE              1.3257     0.2233   5.937 2.91e-09 ***
## same_raceTRUE             0.3603     0.1399   2.575 0.010036 *
## fam_law_enforcementTRUE  -0.5627     0.1622  -3.468 0.000524 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2579.5  on 2294  degrees of freedom
## Residual deviance: 1887.6  on 2287  degrees of freedom
## AIC: 1903.6
##
## Number of Fisher Scoring iterations: 5
```

```
exp(confint(apm_glm))
```

```
## Waiting for profiling to be done...
```

```
##                              2.5 %    97.5 %
## (Intercept)             0.07182462  0.106826
## accusedTRUE             4.57474511 40.090813
## is_blackTRUE            5.06788006  8.815129
## fam_accusedTRUE         4.62782800  8.738251
## death_hesitationTRUE    2.01793799 20.917259
## know_defTRUE            2.43708908  5.854130
## same_raceTRUE           1.08819022  1.883933
## fam_law_enforcementTRUE 0.41209408  0.778917
```

```
oddsRatio_accused <- exp(2.5128)
oddsRatio_is_black<- exp(1.8972 )
oddsRatio_fam_accused <- exp(1.8476)
oddsRatio_death_hesitation <- exp(1.8243)
oddsRatio_know_def <- exp(1.3257)
oddsRatio_same_race <- exp(0.3603)
oddsRatio_fam_law_enforcement <- exp(-0.5627 )

oddsRatio_accused
```

```
## [1] 12.33943
```

```
oddsRatio_is_black
```

```
## [1] 6.6672
```

```
oddsRatio_fam_accused
```

```
## [1] 6.344574
```

```
oddsRatio_death_hesitation
```

```
## [1] 6.198455
```

```
oddsRatio_know_def
```

```
## [1] 3.76482
```

```
oddsRatio_same_race
```

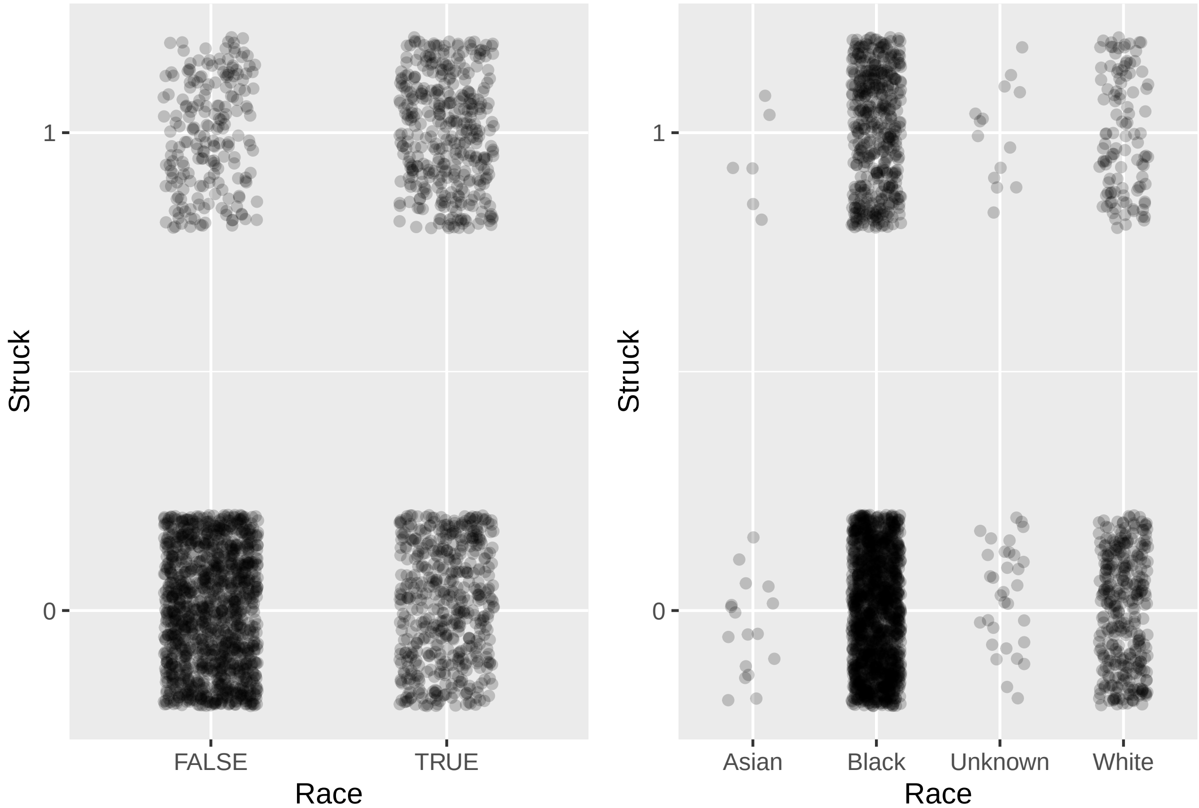
```
## [1] 1.433759
```

```
oddsRatio_fam_law_enforcement
```

```
## [1] 0.5696689
```

We got all of these numbers same or very close to what is in the article!

```
library(ggplot2)
library(patchwork)
sameRace <- ggplot(apm, aes(x = same_race, y = struck_state_bin)) + geom_jitter(width = 0.2, he
defendantRace <- ggplot(apm, aes(x = defendant_race, y = struck_state_bin)) + geom_jitter(widt
sameRace + defendantRace
```




```
sameRace_glm <- glm(struck_state_bin ~ accused + is_black + fam_accused + death_hesitation + kn
defendantRace_glm <- glm(struck_state_bin ~ accused + is_black + fam_accused + death_hesitatio
```

```
nullModel_glm <- glm(struck_state_bin ~ accused + is_black + fam_accused + death_hesitation + k
anova(sameRace_glm, nullModel_glm, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: struck_state_bin ~ accused + is_black + fam_accused + death_hesitation +
##     know_def + same_race + fam_law_enforcement
## Model 2: struck_state_bin ~ accused + is_black + fam_accused + death_hesitation +
##     know_def + fam_law_enforcement
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      2287     1887.5
## 2      2288     1894.1 -1  -6.5328  0.01059 *
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(defendantRace_glm, nullModel_glm)
```

```
## Analysis of Deviance Table
##
## Model 1: struck_state_bin ~ accused + is_black + fam_accused + death_hesitation +
##     know_def + defendant_race + fam_law_enforcement
## Model 2: struck_state_bin ~ accused + is_black + fam_accused + death_hesitation +
##     know_def + fam_law_enforcement
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      2285     1888.1
## 2      2288     1894.1 -3  -5.9644   0.1134
```

```r
AIC(sameRace_glm, defendantRace_glm)
```

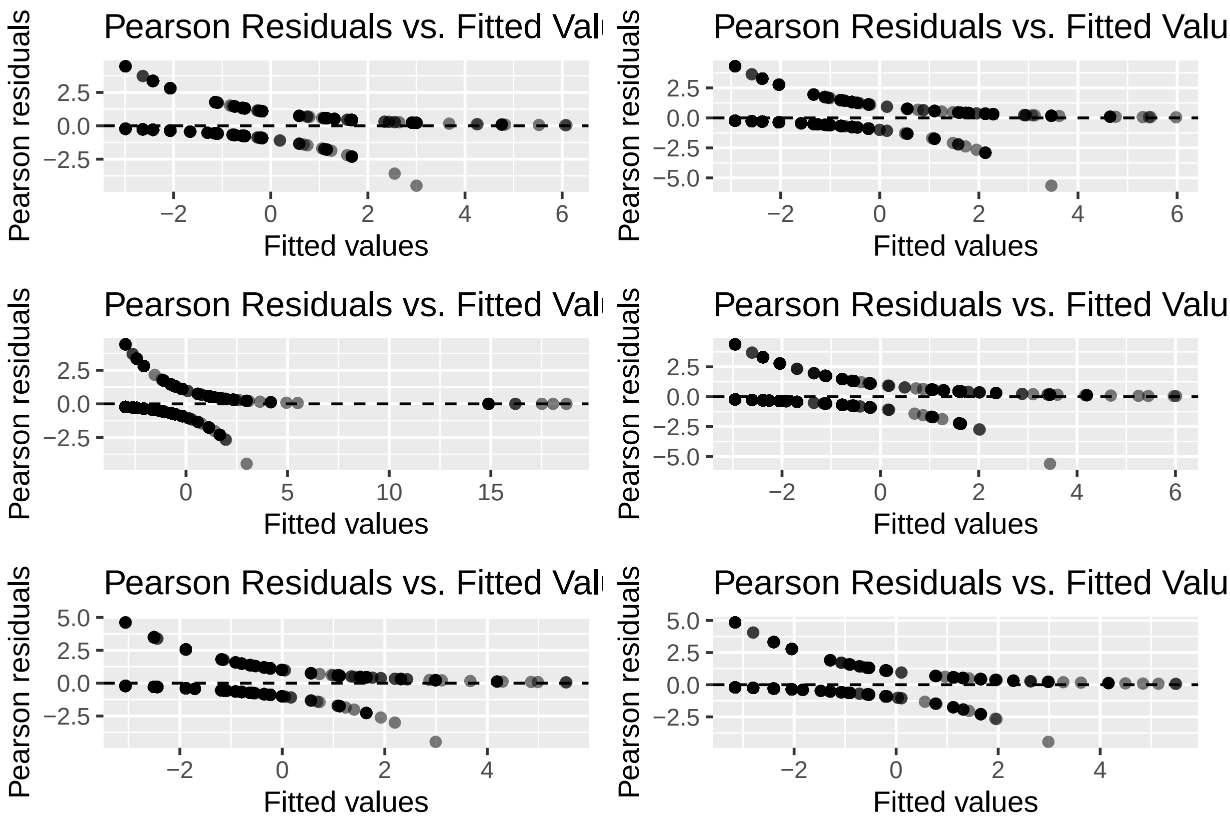
```
##                   df      AIC
## sameRace_glm       8 1903.553
## defendantRace_glm 10 1908.122
```

```r
int_accused <- glm(struck_state_bin ~ is_black * accused + fam_accused + death_hesitation + kno

int_fam_accused <- glm(struck_state_bin ~ is_black * fam_accused + accused + death_hesitation +

int_death_hesitation <- glm(struck_state_bin ~ is_black * death_hesitation + accused + fam_accu
data = apm, family = "binomial")

int_know_def <- glm(struck_state_bin ~ is_black * know_def + accused + fam_accused + death_hes
data = apm, family = "binomial")

int_same_race <- glm(struck_state_bin ~ is_black * same_race + accused + fam_accused + death_he
data = apm, family = "binomial")

int_fam_law <- glm(struck_state_bin ~ is_black * fam_law_enforcement + accused + fam_accused +
data = apm, family = "binomial")
```

```r
library(broom)
library(ggplot2)
library(patchwork)

# 1. is_black * accused
aug_accused <- augment(int_accused, type.residuals = "pearson")
plot1 <- ggplot(aug_accused, aes(.fitted, .resid)) +
geom_point(alpha = 0.5) +
geom_hline(yintercept = 0, linetype = "dashed") +
labs(
x = "Fitted values",
y = "Pearson residuals",
title = "Pearson Residuals vs. Fitted Values: is_black * accused")
```

```
# 2. is_black * fam_accused
aug_fam_accused <- augment(int_fam_accused, type.residuals = "pearson")
plot2 <- ggplot(aug_fam_accused, aes(.fitted, .resid)) +
geom_point(alpha = 0.5) +
geom_hline(yintercept = 0, linetype = "dashed") +
labs(
x = "Fitted values",
y = "Pearson residuals",
title = "Pearson Residuals vs. Fitted Values: is_black * fam_accused")

# 3. is_black * death_hesitation
aug_death_hesitation <- augment(int_death_hesitation, type.residuals = "pearson")
plot3 <- ggplot(aug_death_hesitation, aes(.fitted, .resid)) +
geom_point(alpha = 0.5) +
geom_hline(yintercept = 0, linetype = "dashed") +
labs(
x = "Fitted values",
y = "Pearson residuals",
title = "Pearson Residuals vs. Fitted Values: is_black * death_hesitation")

# 4. is_black * know_def
aug_know_def <- augment(int_know_def, type.residuals = "pearson")
plot4 <- ggplot(aug_know_def, aes(.fitted, .resid)) +
geom_point(alpha = 0.5) +
geom_hline(yintercept = 0, linetype = "dashed") +
labs(
x = "Fitted values",
y = "Pearson residuals",
title = "Pearson Residuals vs. Fitted Values: is_black * know_def")

# 5. is_black * same_race
aug_same_race <- augment(int_same_race, type.residuals = "pearson")
plot5 <- ggplot(aug_same_race, aes(.fitted, .resid)) +
geom_point(alpha = 0.5) +
geom_hline(yintercept = 0, linetype = "dashed") +
labs(x = "Fitted values",y = "Pearson residuals",title = "Pearson Residuals vs. Fitted Values:

# 6. is_black * fam_law_enforcement
aug_fam_law <- augment(int_fam_law, type.residuals = "pearson")
plot6 <- ggplot(aug_fam_law, aes(.fitted, .resid)) +geom_point(alpha = 0.5) +geom_hline(yinterc
x = "Fitted values",
y = "Pearson residuals",
title = "Pearson Residuals vs. Fitted Values: is_black * fam_law_enforcement")


combined_plot <- (plot1 | plot2) /
(plot3 | plot4) /
```

```r
(plot5 | plot6)


combined_plot
```




```r
plot1 <- ggplot(apm, aes(x = factor(is_black), fill = struck_state)) +
geom_bar(position = "fill") +
facet_wrap(~ factor(accused)) +
labs(title = "is_black * accused",
x = "Is Juror Black",
y = "Proportion Struck",
fill = "Struck Status")
```

```r
plot2 <- ggplot(apm, aes(x = factor(is_black), fill = struck_state)) +
geom_bar(position = "fill") +
facet_wrap(~ factor(fam_accused)) +
labs(title = "is_black * fam_accused",
x = "Is Juror Black",
y = "Proportion Struck",
fill = "Struck Status")
```

```r
plot3 <- ggplot(apm, aes(x = factor(is_black), fill = struck_state)) +
geom_bar(position = "fill") +
facet_wrap(~ factor(death_hesitation)) +
labs(title = "is_black * death_hesitation",
```

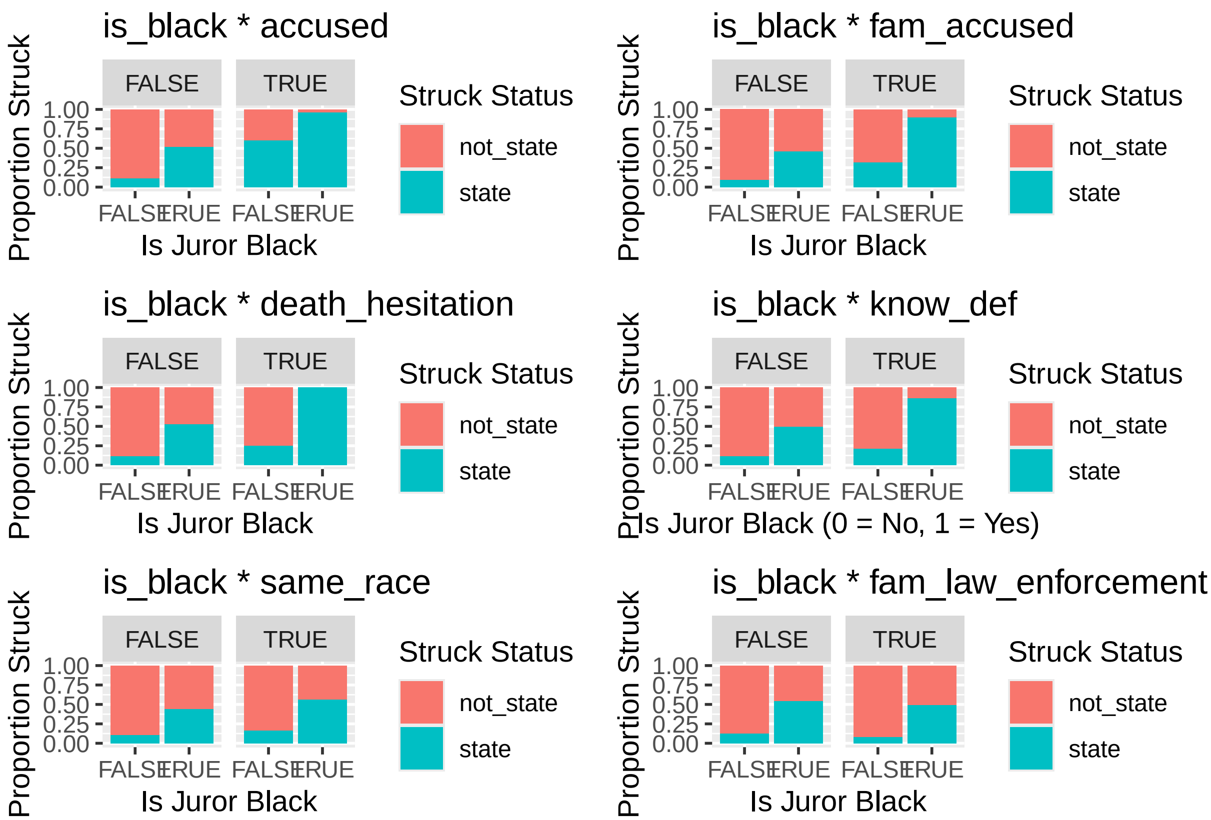
```r
x = "Is Juror Black ",
y = "Proportion Struck",
fill = "Struck Status")
```

```r
plot4 <- ggplot(apm, aes(x = factor(is_black), fill = struck_state)) +
geom_bar(position = "fill") +
facet_wrap(~ factor(know_def)) +
labs(title = "is_black * know_def",
x = "Is Juror Black (0 = No, 1 = Yes)",
y = "Proportion Struck",
fill = "Struck Status")
```

```r
plot5 <- ggplot(apm, aes(x = factor(is_black), fill = struck_state)) +
geom_bar(position = "fill") +
facet_wrap(~ factor(same_race)) +
labs(title = "is_black * same_race",
x = "Is Juror Black",
y = "Proportion Struck",
fill = "Struck Status")
```

```r
plot6 <-ggplot(apm, aes(x = factor(is_black), fill = struck_state)) +
geom_bar(position = "fill") +
facet_wrap(~ factor(fam_law_enforcement)) +
labs(title = "is_black * fam_law_enforcement",
x = "Is Juror Black",
y = "Proportion Struck",
fill = "Struck Status")
```

```r
library(patchwork)

combined_plot <- (plot1 | plot2) /
(plot3 | plot4) /
(plot5 | plot6)

combined_plot
```




```r
anova(nullModel_glm, int_accused, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: struck_state_bin ~ accused + is_black + fam_accused + death_hesitation +
##     know_def + fam_law_enforcement
## Model 2: struck_state_bin ~ is_black * accused + fam_accused + death_hesitation +
##     know_def + same_race + fam_law_enforcement
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      2288     1894.1
## 2      2286     1886.9  2   7.1498  0.02802 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(nullModel_glm, int_fam_accused, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: struck_state_bin ~ accused + is_black + fam_accused + death_hesitation +
##     know_def + fam_law_enforcement
## Model 2: struck_state_bin ~ is_black * fam_accused + accused + death_hesitation +
##     know_def + same_race + fam_law_enforcement
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      2288     1894.1
## 2      2286     1882.9  2    11.23 0.003642 **
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(nullModel_glm, int_death_hesitation, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: struck_state_bin ~ accused + is_black + fam_accused + death_hesitation +
##     know_def + fam_law_enforcement
## Model 2: struck_state_bin ~ is_black * death_hesitation + accused + fam_accused +
##     know_def + same_race + fam_law_enforcement
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      2288     1894.1
## 2      2286     1883.2  2   10.883 0.004332 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(nullModel_glm, int_know_def, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: struck_state_bin ~ accused + is_black + fam_accused + death_hesitation +
##     know_def + fam_law_enforcement
## Model 2: struck_state_bin ~ is_black * know_def + accused + fam_accused +
##     death_hesitation + same_race + fam_law_enforcement
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      2288     1894.1
## 2      2286     1882.2  2   11.932 0.002564 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(nullModel_glm , int_same_race, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: struck_state_bin ~ accused + is_black + fam_accused + death_hesitation +
##     know_def + fam_law_enforcement
## Model 2: struck_state_bin ~ is_black * same_race + accused + fam_accused +
##     death_hesitation + know_def + fam_law_enforcement
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      2288     1894.1
## 2      2286     1884.3  2   9.8183 0.007379 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(nullModel_glm, int_fam_law, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: struck_state_bin ~ accused + is_black + fam_accused + death_hesitation +
##     know_def + fam_law_enforcement
```

```
## Model 2: struck_state_bin ~ is_black * fam_law_enforcement + accused +
##     fam_accused + death_hesitation + know_def + same_race
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      2288     1894.1
## 2      2286     1886.0  2   8.1185  0.01726 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
AIC(nullModel_glm, int_accused)
```

```
##                df      AIC
## nullModel_glm  7 1908.086
## int_accused    9 1904.936
```

```
AIC(nullModel_glm, int_fam_accused)
```

```
##                  df      AIC
## nullModel_glm     7 1908.086
## int_fam_accused  9 1900.856
```

```
AIC(nullModel_glm, int_death_hesitation)
```

```
##                       df      AIC
## nullModel_glm          7 1908.086
## int_death_hesitation  9 1901.203
```

```
AIC(nullModel_glm, int_know_def)
```

```
##                df      AIC
## nullModel_glm  7 1908.086
## int_know_def   9 1900.154
```

```
AIC(nullModel_glm, int_same_race)
```

```
##                df      AIC
## nullModel_glm  7 1908.086
## int_same_race  9 1902.268
```

```
AIC(nullModel_glm, int_fam_law)
```

```
##                df      AIC
## nullModel_glm  7 1908.086
## int_fam_law    9 1903.968
```